Problem Summary

Given a string containing only brackets : ( ) [ ] { } determine if the brackets are valid.

A string is valid if:

→ every opening bracket has a matching closing bracket

→ brackets close in the correct order

Key Idea

Use a stack

→ Push opening brackets onto the stack

→ when we see a closing bracket, check if it matches the most recent opening bracket

Diagram Example

Non valid string : '([])){' Valid string : '([])'

Approach

1) Create a hashmap : closing bracket → matching opening bracket

2) Create a empty stack

3) Traverse each bracket :

→ if it is an opening bracket : push onto stack

→ if it is a closing bracket : check if stack is empty, pop top of stack, compare with expected opening bracket

4) At the end : if stack is empty → valid, otherwise → invalid

Time Complexity : O(n) → each bracket is pushed / popped at most once

Space Complexity : O(n) → worst case all brackets are opening brackets